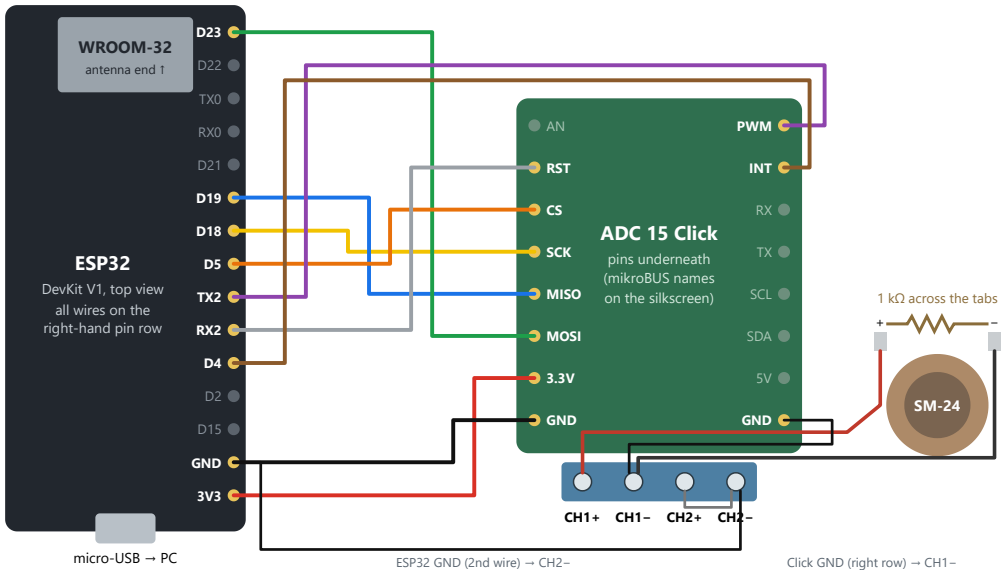
REZON geophone test: SM-24 → ADC 15 Click → ESP32

Bench test · classic ESP32 DevKit V1 (30-pin, WROOM-32) · MikroE ADC 15 Click (ADS131M02 + LTC6903 clock) · SM-24 geophone · powered from the PC over USB · 2026-11-12



A · ESP32 ↔ Click (9 F-F jumpers)

#	Colour	ESP32 pin right row, from the antenna end	Click pin row, from the top (AN / PWM end)	What it does
1	red	**3V3** 15th (last)	**3.3V** left, 7th · mikroBUS 7	Power. **Not 5 V**
2	black	**GND** 14th	**GND** (left row) left, 8th (last) · mikroBUS 8	Ground
3	yellow	**D18** 7th	**SCK** left, 4th · mikroBUS 4	SPI clock
4	blue	**D19** 6th	**MISO** (SDO) left, 5th · mikroBUS 5	Data ADC → ESP32
5	green	**D23** 1st	**MOSI** (SDI) left, 6th · mikroBUS 6	Data ESP32 → ADC
6	orange	**D5** 8th	**CS** left, 3rd · mikroBUS 3	ADC chip select
7	grey	**RX2** (GPIO16) 10th	**RST** left, 2nd · mikroBUS 2	ADC reset
8	purple	**TX2** (GPIO17) 9th	**PWM** right, 1st · mikroBUS 16	Chip select of the clock chip (LTC6903). **Without it the ADC never runs**
9	brown	**D4** 11th	**INT** right, 2nd · mikroBUS 15	DRDY: LOW = new sample

Click pins AN, RX, TX, SCL, SDA, 5V stay empty. The two Click GND pins are joined on the board: the **right GND** (right row, 8th · mikroBUS 9) needs no wire to the ESP32. It is only used as a handy ground for CH1− (wire 13). "Left/right" = Click seen from the component side with AN and PWM at the top; the names printed next to the pins decide.

B · Geophone → Click screw terminals

#	From	To	Note
10	1 kΩ resistor	across the two SM-24 tabs	Damps the coil; solder it right on the tabs
11	SM-24 **+** tab	**CH1+**	One twisted pair from the 6×0.22 cable, about 1 turn per cm
12	SM-24 **−** tab	**CH1−**	
13	Click **GND** (right row)	**CH1−**	Gives the coil a 0 V reference. Use a jumper with one end stripped 5 mm
14	**CH2+**	**CH2−**	Short loop: unused channel shorted
15	ESP32 **GND** (left row, bottom)	**CH2−**	Grounds CH2; the page uses it as the noise floor

Max two wires per screw terminal. If your terminals are labelled AIN0P / AIN0N / AIN1P / AIN1N, then CH1 = AIN0, CH2 = AIN1. If you can't tell + from − on the geophone, it doesn't matter for this test: the trace is just inverted.

> **Check before USB:** 3.3V on the Click goes to **3V3**, never VIN/5V. If the Click has a supply jumper (VCC SEL / 3V3–5V), set it to **3V3**. Stand the geophone upright (spike down) on the table or floor and don't hold it while testing.

Run it

1. Arduino IDE → board **ESP32 Dev Module**. Open `practice/geophone_test/geophone_test.ino`, upload. No libraries needed.
2. Optional: Serial Monitor at **460800** → press **EN**. Expect `ID = 0x22xx` and then lots of `D,…` lines. **Close the Serial Monitor** afterwards.
3. Open `practice/geo_viz/index.html` in **Chrome or Edge** → **Connect ESP32** → pick the COM port.
4. Tap the table 30 cm from the geophone, walk past it, drop a book.

The **Demo** button in the page plays fake footsteps and a fall, so you can try the page before the wiring is done.

If it doesn't work

☐ `ID = 0x0000` or `0xFFFF`: check SPI (wires 3–6), RST (7) and especially **PWM** → **TX2** (8). No clock means no answer.

☐ Page shows 0 samples/s: INT → D4 (wire 9).

☐ CRC errors > 0: wires too long or loose; keep jumpers short.

☐ Flat line while you stomp: wires 11–13, the resistor, and the terminal screws.

☐ "clipping" in the log: lower the ADC gain in the page (×32 → ×8).

☐ Constant 50 Hz hum: wire 13 missing (CH1− not grounded).